

Stephen Murphy

Experienced Mechanical Engineer | Problem Solver | Technical Generalist | MEng

I help businesses deliver products to market through solid engineering design, ambitious goal-setting, effective communication between project teams, and positive work cultures.

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WORK EXPERIENCE

Mechanical Design Manager

SkySafe

01/2023 - 11/2023

San Diego, CA

Accomplishments

- Optimized the mechanical and electrical design of 2 drone detection platforms for improved ease of manufacture, cost, and reliability
- Led and performed technical work including Finite Element Analysis (thermal, and structural), CFD, drawing reviews (ASME Y14.5-2009), compliance (RoHS & CE), and cross-functional communication
- Collaborated heavily with manufacturing to institute line quality checks, test protocols, build tracking, and supplier qualification

Mechanical Design Manager

SharkNinja

03/2021 - 09/2022

Boston, MA

Accomplishments

- Owned the product design engineering of Shark FlexStyle, HyperAir, SpeedStyle, and SmoothStyle high volume consumer electronics products from PoC to mass production at volumes > 100k units/year
- Led and performed mechanical design work including CAD (CREO), electronics integration, injection molding, DFM, CFD, FEA, DFMEA, and tolerance analysis
- Promoted to US Personal Care Engineering Lead, where I led a team of 4 AD/NPD engineers and cross-functional representation

Software Engineering Manager, Software Engineer

Intel Corporation, Drone Group

09/2018 - 01/2020

St. Leon-Rot, Germany

Accomplishments

- Programmed UIs in Java, JavaFX, and CSS
- Led a 9-person software team to the successful release of Intel Mission Control, a drone flight planning application for Windows

Senior Mechanical Engineer

Intel Corporation, Drone Group

10/2016 - 09/2018

St. Leon-Rot, Germany

Accomplishments

- Successfully designed a commercial UAS from scratch including a 2kg aircraft, batteries, GNSS base station, remote, and charger, resulting in my promotion to lead a team of 9 mechanical engineers
- Designed and prototyped composite, machined, injection molded, sheet metal, and laser cut parts using CAD/FEA and Star-CCM+ CFD

Mechanical/Aeronautical Engineer

MAVinci

10/2015 - 10/2016

St. Leon-Rot, BW, Germany

Achievements/Tasks

- Designed and built a custom 3-axis CNC and a 2kg fixed wing UAS

STRENGTHS

Industries

Aerospace, consumer electronics, telecom, metal additive manufacturing, drones, batteries

CAD

SOLIDWORKS, CREO, Fusion 360

Programming & Simulation

Java, Python, MATLAB, Simulink, Abaqus FEA, Star-CCM+ CFD, COMSOL Multiphysics FEA & CFD

Mechanical Design

Propulsion, DFM, consumer products, batteries, injection molding, motors, heaters, machine design, thermal management, acoustics, sheet metal, machining, DFMEA, tolerance analysis

Leadership

Building end-to-end engineering teams, system architecture, technical ownership, engineering management, creating engaging and high performing cultures, executive communication

EDUCATION

Master of Engineering in Mechanical Engineering

University of Victoria

01/2020 - 12/2020

Masters project: Thermo-electrochemical Model of a Zinc-air Flow Battery (written in Python)

Bachelor of Engineering in Mechanical Engineering (with Co-op)

University of Victoria

09/2010 - 06/2015

Top 15% of class; Honor's thesis on HALE biplane design

LANGUAGES

English

Native Speaker

German

Limited Working Proficiency

PERSONAL INTERESTS

Cooking while listening to old soul

Building road bikes

Ultralight hiking